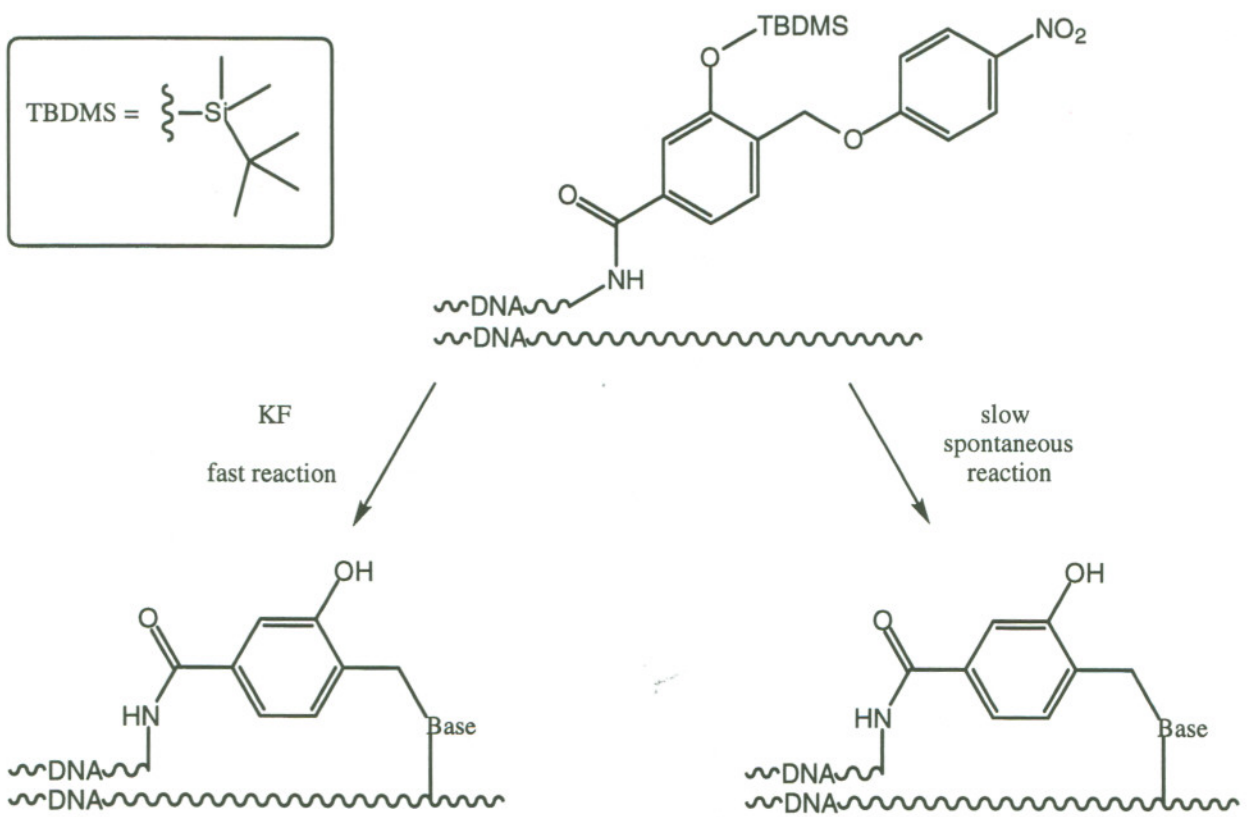
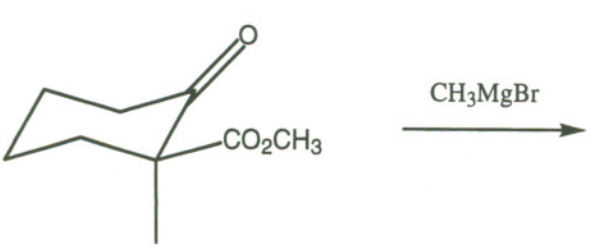
Organic Cumulative Exam **October 22, 1999**

1) Prof. Steve Rokita (University of Maryland) discussed his work with DNA alkylating reagents. According to the scheme below, he noted that there were two ways to activate the alkylating reagent for alkylation of the complementary DNA target strand: (1) by addition of KF, and (2) by association with duplex DNA. Addition of KF was seen to promote rapid crosslinking with the complementary DNA target strand as anticipated. However, it was also observed that simple association of the complementary DNA strand with the DNA strand conjugated to the alkylating reagent spontaneously produced crosslinked complexes in a slower reaction. All evidence suggested that both reactions occurred through the same reactive intermediate.

a) Give the most likely mechanism for the crosslinking reaction in the presence of KF. The nucleic acid base which participates in the crosslinking reaction can be represented as a generic nucleophile (:Nu-H) in your mechanism.

b) Give a rational mechanism for the crosslinking reaction in the absence of KF.

c) Based on your proposed mechanism for the spontaneous crosslinking in the absence of KF, explain why Prof. Rokita might have observed diminished crosslinking under higher NaCl conditions.



2) Show the expected major diastereomer from the following reaction and provide two mechanistic reasons for its favored formation.